

Write a brief conclusion explaining what you have learned about:

1. What is an HPC cluster?
2. What is a compute node?
3. What is a CPU socket?
4. What is cache locality?
5. What is NUMA?
6. What is OpenMP?
7. What is MPI?
8. What is an MPI rank?
9. What is a thread?
10. What is parallel scalability?
11. What is communication latency?
12. What is memory bandwidth?
13. What is FLOPS?
14. What is R_{max} ?
15. What is R_{peak} ?
16. What is InfiniBand?
17. What is RDMA?
18. What is a GPU accelerator?
19. What is heterogeneous computing?
20. Why is workload characterization important in HPC?

Write a brief conclusion explaining what you have learned about:

1. HPC hardware architecture
2. CPU and memory hierarchy
3. UMA and NUMA
4. Shared and distributed memory programming
5. OpenMP and MPI
6. Parallel performance
7. FLOPS and scalability
8. Network interconnects
9. Modern CPU-GPU HPC systems